Statistical Analysis Report (NYC Yellow Taxi Data)

Problem Statement

Our goal is to use data driven insights to maximise revenue streams for taxi drivers in order to meet this need. Our research aims to determine whether payment method have an impact on fare pricing by focusing on the relationship between payment type and fare amount.

Research Question

Is there a relationship between total fare amount and payment type?
Can we nudge customers towards payment methods that generate higher revenue for drivers, without negatively impacting customer experience?

Objective of the Analysis

The primary objective of this analysis is to understand:

- Customer payment behavior (cash vs card)
- Trip characteristics such as fare amount, distance, and duration
- Relationship between payment type and fare patterns
- Statistical validation of differences in fare between payment methods

Dataset Overview

- **Dataset Name:** NYC Yellow Taxi Trip Data
- **Columns used for this research:**
  - Passenger count
  - Payment type
  - Fare amount
  - Trip distance
  - Pickup & drop-off datetime

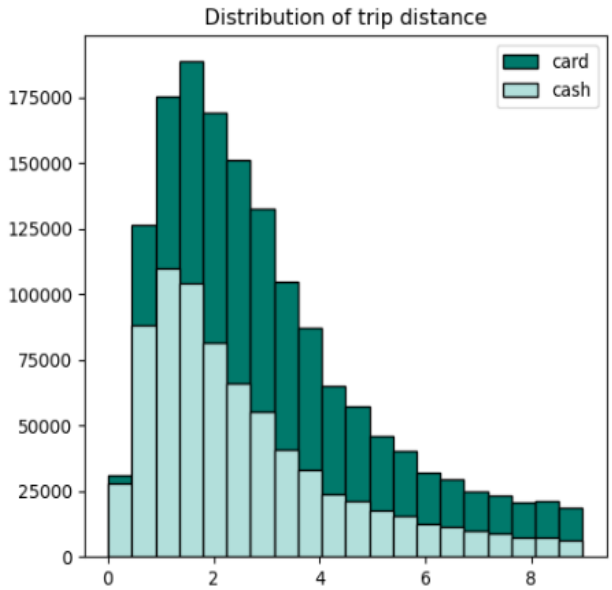
passenger_count	payment_type	fare_amount	trip_distance	duration
1	card	6.0	1.20	4.800000
1	card	7.0	1.20	7.416667
1	card	6.0	0.60	6.183333
1	card	5.5	0.80	4.850000
1	cash	2.5	0.03	0.883333
...	...	...	...	...
4	card	10.5	2.40	12.383333
3	card	10.0	2.09	14.800000
1	card	17.5	4.11	21.500000
1	card	13.0	2.13	19.000000
1	card	12.5	2.55	16.283333

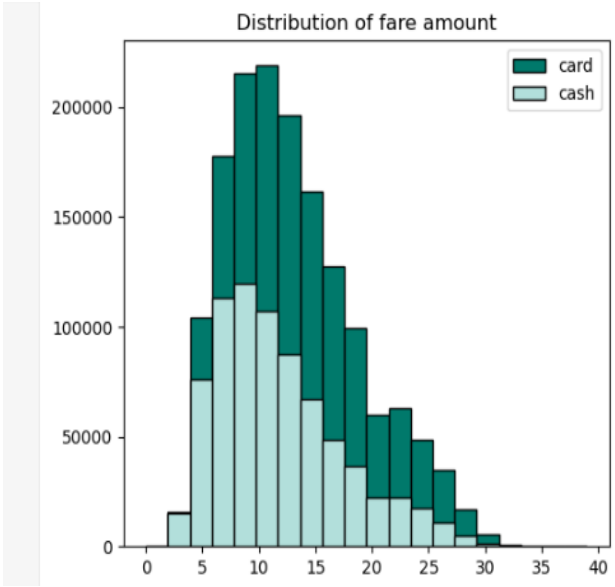
Methodology

Step	Description
Descriptive Analysis	Performed statistical analysis to summarize key aspects of the data, focusing on fare amounts and payment types
Hypothesis Testing	Conducted a T-test to evaluate the relationship between payment type and fare amount, testing the hypothesis that different payment methods influence fare amount

Insights

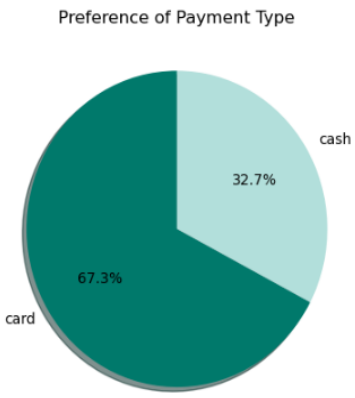
- Customers paying with cards tend to have a slightly higher average trip distance and fare amount compare to those paying with cash
- Indicates that customers prefer to pay more with card when they have high fare amount and long trip distance





Preference of Payment Types

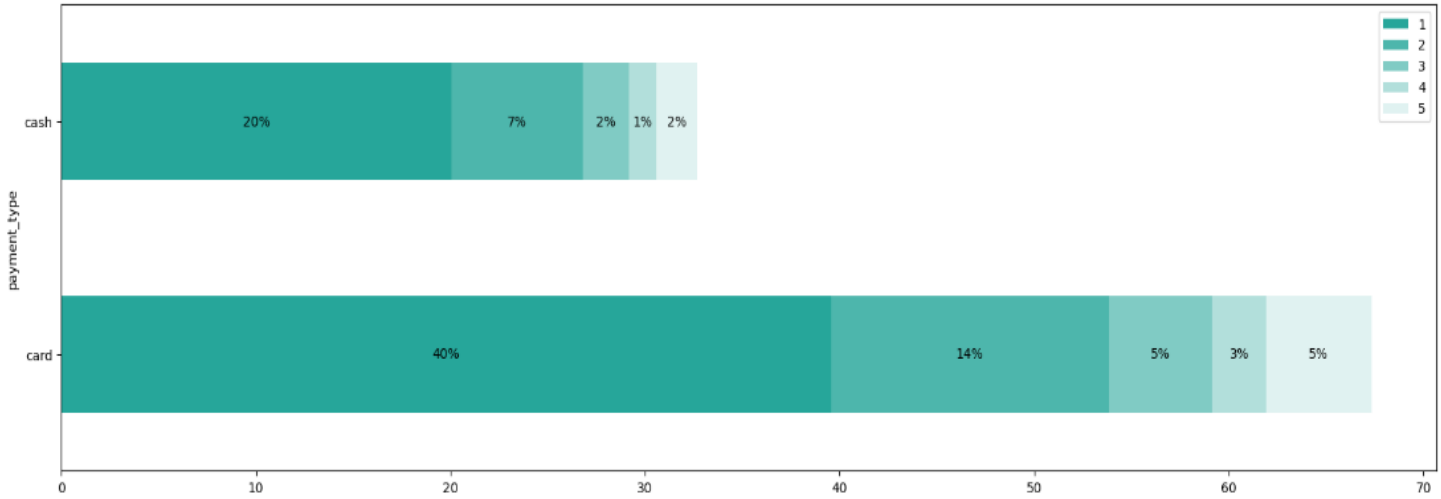
- The proportion of customers paying with cards is higher than those paying with cash, with card payments accounting for 67.3% of all transactions compared to cash payments at 32.7%.
- This indicates a strong preference among customers for using card payments over cash, potentially due to convenience, security, or incentives offered for card transactions.



Passenger count analysis

- Among card payments, rides with a single passenger (passenger_count = 1) comprise the largest proportion, constituting 40% of all card transactions.
- Similarly, cash payments are predominantly associated with single passenger rides, making up 20% of all cash transactions.
- There Is a noticeable decrease in the percentage of transactions as the passenger count increases, suggesting that larger groups are less likely to use taxis or may opt for alternative payment methods.

- These insights emphasize the importance of considering both payment method and passenger count when analyzing transaction data, as they provide valuable insights into customer behavior and preferences.



Hypothesis Testing

- **Null Hypothesis (H_0):** There is no difference in average fare between card and cash users.
- **Alternative Hypothesis (H_1):** There is a difference in average fare between card and cash users

With a T-statistics of 169.21 and a p-value of less than 0.05, we reject the null hypothesis, suggesting that there is a difference in average fare between the two payment methods.

Recommendations

1. **Promote Digital Payments**
   - Offer discounts for card/online payments
   - Improves revenue and convenience
2. **Target High-Value Customers**
   - Focus marketing on card users
   - Provide loyalty programs
3. **Optimize Pricing Strategy**
   - Analyze short vs long trip behavior
   - Introduce dynamic pricing if needed
4. **Improve Customer Experience**
   - Enhance digital payment systems
   - Reduce friction in payment process

Conclusion

This analysis clearly shows that:

- Payment type plays a significant role in customer behavior
- Card users generate higher revenue compared to cash users
- Statistical testing confirms that the difference is not random